Lobster Simulator Game

Session 14

Strategies for competing in Industries & Markets

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Agenda for today

1. Instructions for the game
2. Play the Lobster game
3. Underlying theory of the game


MENORCA


Mallorca




IBIZA


FORMENTERA

- **Three firms** (***Alpha***, ***Bravo*** and ***Charlie***) and Each firm has one boat that operates in one of the **four Balearic Island** (***Mallorca***, ***Menorca***, ***Ibiza*** or ***Formentera***) → 3 teams & 4 sub-teams per group.



	Alpha	Bravo	Charlie
Mallorca	Group 1.a & 6.a	Group 3.a & 8.a	Group 5.a & 10.a
Menorca	Group 1.b & 6.b	Group 3.b & 8.b	Group 5.b & 10.b
Ibiza	Group 2.a & 7.a	Group 4.a & 9.a	Group 5.c & 10.c
Formentera	Group 2.b & 7.b	Group 4.b & 9.b	Group 5.d & 10.d

Please check: Which firm are you working for? And on which island? Find your fellow fisher

*Boats from the same island **cannot** "sit" next to each other.*

So please seat by Firms (people of the same firm seat together)

Instruction in brief (1/3)

- Each team tries to **catch the amount of lobsters that maximizes its profits**. Your boat has to decide how many pounds to catch and bring to the market:

→ **Each team can´t catch more than 75 lobsters per day**

- There will be **10 rounds** (days) of fishing
- **NO COLLUSION** → **Tacit and predetermined collusion** with other boats **is penalized** → out of the market for some days …

Instruction in brief (2/3)

- Once all decisions have been made, each boat's representative has to **bring the announcement sheet** to the market (= front desk) with the agreed-upon quantity of lobsters
- Information about each firm's production will then become public knowledge
- Goal: **Maximize your profits (not outperform the other firms)!**

Instruction in brief (3/3) → Total rounds 9

Only decide on Quantity (Price depends on Quantity of all the three firms)

1. Rounds 1, 2 & 3 → **Cournot NE under Asymmetry of Information**

Each boat makes its decision without knowing anything about the others groups' decisions.

2. Rounds 4, 5 & 6 → **Cournot NE under Fist Mover Advantage (FMA)**

Alpha (for each island) announces their quantity and the other companies (Bravo & Charlie for the other islands) decide simultaneously and secretly. Then the other firms do the same on rounds 5 & 6.

3. Rounds 7, 8 & 9 → **Cournot NE under Signaling**

Alpha (for each island) announce the quantity. This is a signaling not FMA and therefore the company can signal but it might deviate. Then the other firms do the same on rounds 8 & 9.

4. Round 10 → **Collusion agreement**

All the firms talks and collude to reach an agreement on quantities.

The numbers in brief

- Each boat incurs costs of **$5.00** per pound of lobster (this includes the opportunity cost of time)
- The **market price** of lobsters is determined by this equation:

$$P(Qa, Qb, Qc) = 45 - 0.2 (Qa + Qb + Qc)$$

- The **profit** of each firm equals the number of pounds caught multiplied by its profit margin, namely:

$$PROFITS (Qa, Qb, Qc) = Qa [P(Qa, Qb, Qc) - 5]$$

Questions?

MALLORCA								
ROUND	QA	QB	QC	PRICE	PROF A	PROF B	PROF C	TOTAL
1	50.00	75.00	50.00	10.00	250.00	375.00	250.00	875.00
2	75.00	60.00	45.00	9.00	300.00	240.00	180.00	720.00
3	30.00	65.00	55.00	15.00	300.00	650.00	550.00	1500.00
4	34.00	64.00	30.00	19.40	489.60	921.60	432.00	1843.20
5	34.00	55.00	50.00	17.20	414.80	671.00	610.00	1695.80
6	65.00	59.00	75.00	5.20	13.00	11.80	15.00	39.80
7				45.00	0.00	0.00	0.00	0.00
8				45.00	0.00	0.00	0.00	0.00
9				45.00	0.00	0.00	0.00	0.00
10				45.00	0.00	0.00	0.00	0.00
11				45.00	0.00	0.00	0.00	0.00
12				45.00	0.00	0.00	0.00	0.00
Total					1767.40	2869.40	2037.00	6673.80

MENORCA								
ROUND	QA	QB	QC	PRICE	PROF A	PROF B	PROF C	TOTAL
1	65.00	45.00	30.00	17.00	780.00	540.00	360.00	1680.00
2	50.00	55.00	55.00	13.00	400.00	440.00	440.00	1280.00
3	50.00	44.00	50.00	16.20	560.00	492.80	560.00	1612.80
4	47.00	45.00	47.00	17.20	573.40	549.00	573.40	1695.80
5	47.00	47.00	47.00	16.80	554.60	554.60	554.60	1663.80
6	47.00	59.00	60.00	11.80	319.60	401.20	408.00	1128.80
7				45.00	0.00	0.00	0.00	0.00
8				45.00	0.00	0.00	0.00	0.00
9				45.00	0.00	0.00	0.00	0.00
10				45.00	0.00	0.00	0.00	0.00
11				45.00	0.00	0.00	0.00	0.00
12				45.00	0.00	0.00	0.00	0.00
Total					3187.60	2977.60	2896.00	9061.20

IBIZA								
ROUND	QA	QB	QC	PRICE	PROF A	PROF B	PROF C	TOTAL
1	50.00	45.00	45.00	17.00	600.00	540.00	540.00	1680.00
2	75.00	50.00	55.00	9.00	300.00	200.00	220.00	720.00
3	37.00	55.00	55.00	15.60	392.20	583.00	583.00	1558.20
4	50.00	45.00	45.00	17.00	600.00	540.00	540.00	1680.00
5	45.00	47.50	45.00	17.50	562.50	593.75	562.50	1718.75
6	48.00	49.00	50.00	15.60	508.80	519.40	530.00	1558.20
7				45.00	0.00	0.00	0.00	0.00
8				45.00	0.00	0.00	0.00	0.00
9				45.00	0.00	0.00	0.00	0.00
10				45.00	0.00	0.00	0.00	0.00
11				45.00	0.00	0.00	0.00	0.00
12				45.00	0.00	0.00	0.00	0.00
Total					2963.50	2976.15	2975.50	8915.15

FORMENTERA								
ROUND	QA	QB	QC	PRICE	PROF A	PROF B	PROF C	TOTAL
1	67.00	35.00	70.00	10.60	375.20	196.00	392.00	963.20
2	67.00	60.00	48.00	10.00	335.00	300.00	240.00	875.00
3	67.00	75.00	48.00	7.00	134.00	150.00	96.00	380.00
4	75.00	70.00	50.00	6.00	75.00	70.00	50.00	195.00
5	67.00	70.00	75.00	2.60	-160.80	-168.00	-180.00	-508.80
6	67.00	67.00	75.00	3.20	-120.60	-120.60	-135.00	-376.20
7				45.00	0.00	0.00	0.00	0.00
8				45.00	0.00	0.00	0.00	0.00
9				45.00	0.00	0.00	0.00	0.00
10				45.00	0.00	0.00	0.00	0.00
11				45.00	0.00	0.00	0.00	0.00
12				45.00	0.00	0.00	0.00	0.00
Total					637.80	427.40	463.00	1528.20

And the winner is …
Why have you won?

- Because of your moves?
- Your competitors' moves?
- The 'Island's Setting'? (other external conditions)
- Did First Mover Advantage help?
- Was signalling an efficient strategy?

Underlying theory of the game

- Market in which the actions of individual firms materially affect the overall market (and thus other firms' profits) → oligopoly

[Reminder: In perfectly competitive and monopolistically competitive markets, sellers do not believe that their price or output will affect rivals' prices or output]

- Lobster game:
  - 3 firms, identical product, forced to charge the same price
  - only decide on quantity, price = price that "clears the market" given the total produced quantity



- **Cournot NE under Asymmetry of Information**
- **Cournot NE with First Mover Advantage**
- **Cournot NE with Signalling**

Cournot Equilibrium quantity and price

- Equilibrium output for each firm: 50 pounds
- Total output in equilibrium: 150 pounds
- Equilibrium price: $15
- Equilibrium profit for each firm: $500 Profit
- Profit for all the firms $1500.

Monopoly Cournot quantity and price

- Monopoly output for monopolistic firm: 100 lobsters

  Other firms: 0 pounds
- Monopoly price: $25
- Monopoly profit for monopolistic firm: $2000

Best Cooperative quantity and price

- Monopoly output for monopolistic firm: 100 pounds
- **Q (A) = Q (B) = Q (C) = 100 /3 = 33 lobsters per firm**

Perfect Competition Cournot quantity and price

- Perfect Competition output (All the firms): 200 pounds
- Perfect Competition price: $ 5
- Perfect Competition profit: $0 = price – cost = $5 – $5

P(Qa, Qb, Qc) = 45 - 0.2 (Qa + Qb + Qc) →

Price that makes benefit = 0 is $5 – $5 = $0

Which (Qa + Qb + Qc) ? → 5 = 45 – 0.2 (Qa + Qb + Qc)

5 + 0.2 (Qa + Qb + Qc) = 45 → 0.2 (Qa + Qb + Qc) = 200 Lobsters

P (Qa, Qb, Qc) = 45 - 0.2 (Qa + Qb + Qc)

PROFITS (Qa, Qb, Qc) = Qa [P(Qa, Qb, Qc) - 5]

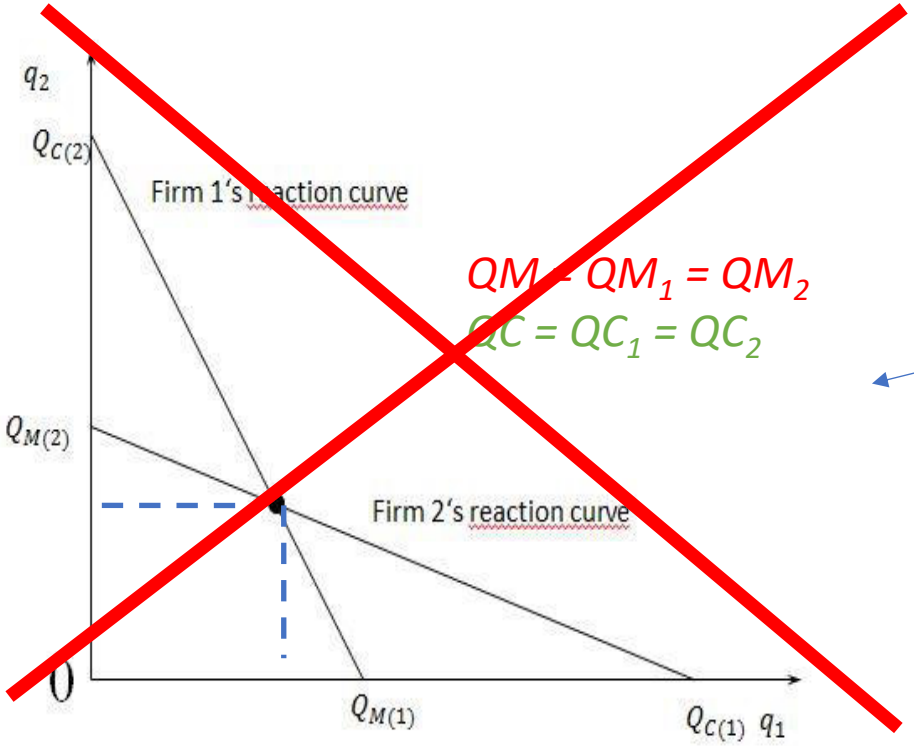
$$Q_D = q_1^* + q_2^*$$

$$Q_M < Q_D < Q_C$$

Monopoly **33 u.**

Cournot **50 u.**

Competition **200 u.**



Three Firms → 3 dimensions.

Not applicable this graph. Only for a two dimensions Cournout Model

Observations (1/2)

- **Optimal decision (and profit) depend on the actions of the other firms**
- **What would be a agreement on quantity among the 3 boats yielding higher profits?**
  - Producing monopoly quantity → Q (a) = Q (b) = Q (c) = 33.33 lobsters
  - Profit (a) = Profit (b) = Profit (c) = $ 666.6
- **Deterring vs. Accommodating**
  - Aggressive behaviour is penalized
  - Deterring strategy does not lead to the Nash Equilibrium → Incentivises other firms to deviate
- **The Firm with the highest profit: Most adapting firm**
  - Why? → Best Reaction Function
  - Self-interest vs. group interest? Cooperation vs. adaption?

Observations (2/2)

- "**Out of Equilibrium" scenarios**: Firms have an incentive to unilaterally deviate
  - Firms adjust their quantities and move from the 'out of equilibrium' scenario
  - The adjustment process continues over an iterated game until the Equilibrium quantity is reached

- **The Cournot equilibrium industry output does not maximize industry profits** (i.e. it is not pareto optimal; max. for monopoly quantity and price)
- By maximizing their own profit independently, in Cournot, firms produce more output than if they colluded to maximize profits
- In the Cournot equilibrium, the market price (and profits) reduces if one of the firms expands its production (unilaterally deviates)

Revenue Destruction Effect

- Deviations provoke the so called Revenue Destruction Effect
- Small firms have higher incentive to deviate and disrupt price stability, because they have a smaller revenue destruction
- With the number of firms increasing (n higher), the Cournot prices fall (p lower)
- Cournot model is often used to plan market reaction in presence of merges and acquisition by the Antitrust authorities

First mover advantage

- In the game: “moving first” generates an advantage
- More generally (beyond games) early mover advantages are typically:
  - Learning Curve
  - Reputation and Buyer Uncertainty
  - Switching Costs
- But being the first mover can be a disadvantage as well:
  - Lack of understanding
  - Failure of the technology
  - Lack of complementary assets

First mover advantage

- Going from simultaneous to sequential, the game has no order advantage if it has a dominant strategy for both players (i.e., the order does not matter)

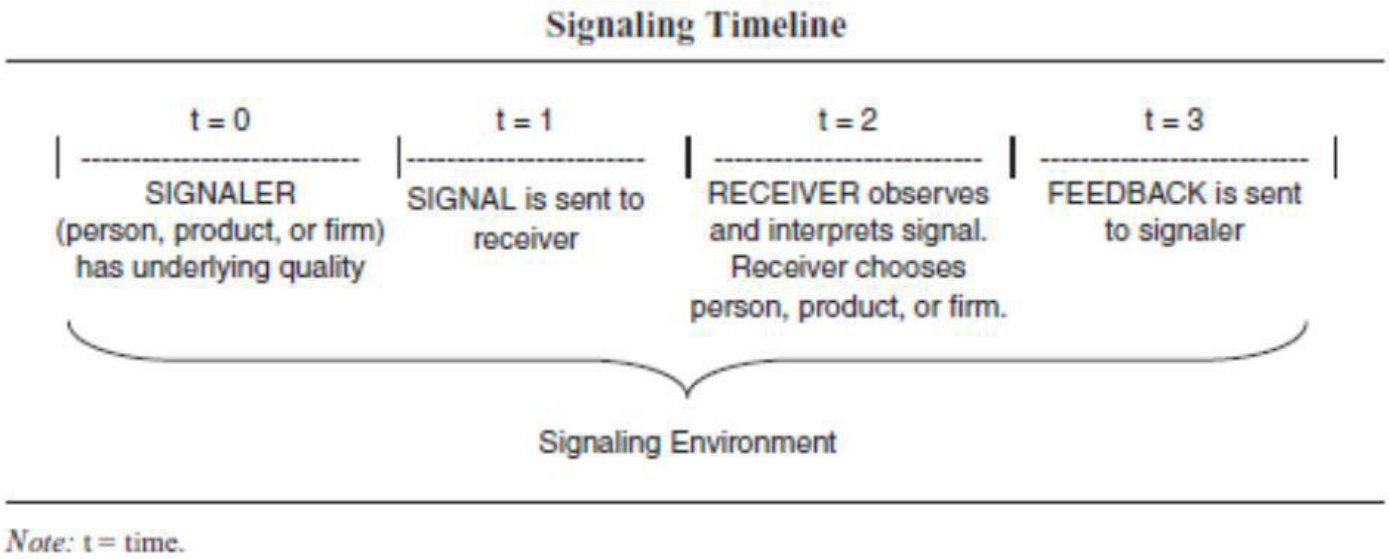
- The incumbent can transform a simultaneous game back to a sequential game by making a strategic commitment move .

Asymmetry of Information and Signaling Theory

- **Signaling theory** is useful for describing behavior when two parties (individuals or organizations) have access to different information.
  - Typically, one party, the sender, must choose whether and how to **communicate (or signal) that information**, and the other party, the receiver, must choose how to interpret the signal.
  - Signaling theory holds a prominent position in a variety of management literatures, including strategic management, entrepreneurship, and human resource management.
- George Akerlof, Michael Spence, and Joseph Stiglitz received the 2001 Nobel Prize in Economics for their work in information economics.

Asymmetry of Information and Signaling Theory

- The signal could create commitment and if the feedback is positive, it also creates trust
  - Collusive Strategies
  - Dynamic Models to sustain NE over time



Note: t = time.

Dynamic vs. Static Models

- Dynamic models can address questions that static models cannot (Example: What determines the intensity of price

- What appears as short term profits (in a static model) are often followed by long term negative effects (in a dynamic model)

- Static models cannot explain how firms can maintain prices above competitive levels without formal collusion

Thanks a lot!

Any questions?

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